



NEUROSCIENCE

Macroscale connectome topographical structure reveals the biomechanisms of brain dysfunction in Alzheimer's disease

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The intricate spatial configurations of brain networks offer essential insights into understanding the specific patterns of brain abnormalities and the underlying biological mechanisms associated with Alzheimer's disease (AD), normal aging, and other neurodegenerative disorders. This study investigated alterations in the topographical structure of the brain related to aging and neurodegenerative diseases by analyzing brain gradients derived from structural MRI data across multiple cohorts ($n = 7323$). The analysis identified distinct gradient patterns in AD, aging, and other neurodegenerative conditions. Gene enrichment analysis indicated that inorganic ion transmembrane transport was the most significant term in normal aging, while chemical synaptic transmission is a common enrichment term across various neurodegenerative diseases. Moreover, the findings show that each disorder exhibits unique dysfunctional neurophysiological characteristics. These insights are pivotal for elucidating the distinct biological mechanisms underlying AD, thereby enhancing our understanding of its unique clinical phenotypes in contrast to normal aging and other neurodegenerative disorders.

INTRODUCTION

Alzheimer's disease (AD) is a prevalent neurodegenerative disorder that exhibits neuron loss, resulting in structural atrophy and cognitive decline (1, 2). Understanding the biomechanisms underlying the structural changes associated with cognitive decline in AD is of great importance in unraveling the complex biological processes of the disease and informing treatment strategies (3). Its shared pathophysiologies and clinical phenotypes with normal aging, frontotemporal dementia (FTD), Parkinson's disease (PD), and other neurodegenerative disorders underscore the challenges in deciphering the biomechanisms underlying these structural changes and their cognitive implications (4).

The human brain is a complex entity consisting of billions of interconnected neurons, forming an intricate network of connections (5). The classical connectome theory posits that the brain operates as a complicated network, facilitating the transformation and integration of information. This framework serves as a fundamental model for elucidating the complexities of brain cognitive functions (6, 7). A key

assumption in neuroscience is that the topographical structure of the brain network provides an organizing principle that constrains its cognitive processes (8–10). Meanwhile, the dysfunction of those hierarchical structures of regional variations represents the imbalance of brain segregation and integration, leading to deficits in cognition (11, 12).

Recent advancements in neuroscience have introduced a novel neuroimaging measure known as the “brain gradient,” which represents the hierarchical organizing axes describing gradual transitions in structural characteristics (8), and provides a framework for understanding the intricate spatial distribution of hierarchical organization based on macroscale brain network (13). Recent multimodal studies have uncovered the existence of a brain gradient extending from primary sensory regions to transmodal association areas. This gradient is linked to increasing representational abstraction along the cortical hierarchy and is correlated to microscale gradients in cytoarchitecture and gene expression profiles (14), underscoring its biological plausibility. Moreover, the investigation of brain gradients can potentially enhance our understanding of the brain changes in prevalent neurological and neuropsychiatric conditions (15–18). Consequently, the concept of the brain gradient provides insights into the biological mechanisms underpinning cognitive impairments in AD. However, the underlying organizational structure of the human brain and what changes in AD and its biological basis remain incompletely elucidated.

The brain gradients were generally accessed on the basis of the macroscale brain network, i.e., cortical morphology (11, 13, 19, 20), microstructure (21, 22), and functional network (8, 23). The dynamics of numerous natural systems are fundamentally shaped and limited by their underlying structural characteristics. The brain is no exception to this rule, as its anatomy inherently constrains and influences its functional processes (24). Structural similarity is a growing focus for magnetic resonance imaging (MRI) of connectomes and provides insight for a better understanding of healthy and disordered brain development (25). On the basis of these insights, we have formulated a hypothesis that the hierarchical organization of macroscale brain

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morphology has a constraining influence on cortical function and is closely associated with cognitive decline in AD (26). A regional radiomics similarity network (R2SN) is a macroscale network based on the similarity of the radiomics features, with robustness, stability, and biological basis (27). R2SN can reflect the imperceptible changes of the brain and provide a perspective for understanding the human brain based on structural T1-weighted MRI data. Consequently, the R2SN gradient holds considerable potential for investigating brain mechanisms underlying cognition and identifying specific dysfunctions of the hierarchical structures associated with neurodegenerative disease (28).

The primary aim of this study is to investigate the specific altered topographical structure of the brain and its biological basis in AD. For these purposes, we first introduced a macroscale brain network, the R2SN, which enables the capture of morphological similarities among different brain regions (27). Subsequently, we computed the principal gradient to characterize the topographical organization of the brain. Furthermore, we examined the altered gradient and the associated biological mechanisms in AD. Crucially, our investigation delineated the distinct alterations in gradients and transcriptional profiles, comparing AD with aging and other forms of dementia in terms of both similarity and specificity (Fig. 1). This study introduces a biologically

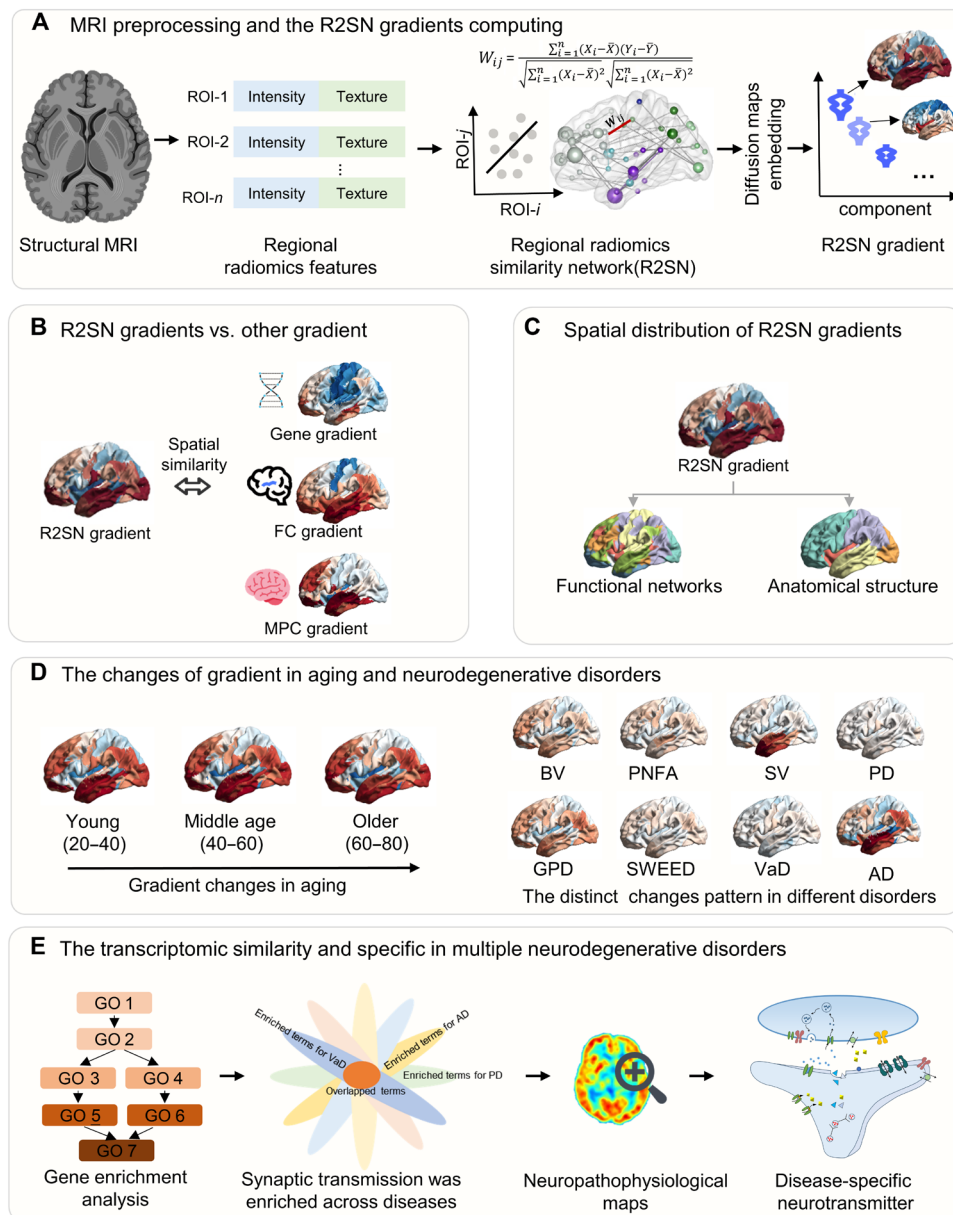


Fig. 1. The pipeline of the present study. (A) The construction of the regional radiomics similarity network (R2SN) and computing R2SN gradient; (B) the spatial similarity between the R2SN gradient and genetics gradient, functional connectome (FC) gradient, and microstructure profile covariance (MPC) gradient; (C) the spatial distribution of R2SN in Yeo functional networks and anatomical structure; (D) the distinct altered pattern of the R2SN gradient in aging, AD and other neurodegenerative disorders; (E) the similarity enriched gene terms and the specific neurotransmitters across neurodegenerative disorders.

plausible framework to elucidate the topographical structure of the brain, offering valuable insights into the overlapping and distinct dysfunctions observed in AD, aging, and other neurodegenerative disorders.

RESULTS

R2SN gradient is biologically plausible for capturing distinct patterns among specific brain parcellation

A total of 7323 participants were included in the present study (Table 1). First, the R2SN gradient exhibited remarkable reliability and consistency across diverse machine parameters, with intraclass correlation coefficient (ICC) values exceeding 0.7 among 90% of brain regions in two traveling datasets (fig. S1). This observation underscores the robustness and transferability of the R2SN gradient, even when confronted with differing MRI machine parameters.

In young adults, the group-level R2SN gradient demonstrates significant correlations with genetic gradients ($R = 0.39$ and $P = 8.78 \times 10^{-10}$), functional gradients ($R = 0.28$ and $P = 3.02 \times 10^{-5}$), and microstructure profile covariance (MPC) gradients ($R = 0.19$ and $P = 0.005$) (Fig. 2A). Furthermore, the R2SN gradient exhibits notable spatial heterogeneity among Yeo networks. Specifically, the lowest R2SN gradient is observed in the unimodal region, encompassing the visual network (VIS), somatomotor network (SM), dorsal attention network (DAN), and ventral attention network (VAN). In contrast, the higher R2SN gradient is localized in the limbic network (LIMB), frontoparietal network (FPN), and default network (DMN) ($P < 0.001$ and $T < -28.35$) (Fig. 2B). Therefore, the R2SN gradient reveals a continuum with a hierarchical structure from the unimodal

region to the transmodal region. In addition, the R2SN gradient shows distinct spatial heterogeneity across anatomical parcellation (Fig. 2C). Besides, the low second R2SN gradient is observed in the transmodal region, such as DMN and LIMB, and the high second R2SN gradient is localized in the unimodal region, including SM and DAN (fig. S2). We have demonstrated the specific relationships between brain gradients and age or sex (fig. S3), suggesting that the brain gradients are influenced by various covariates, each leading to particular alterations.

R2SN gradient of AD was correlated with the neuronal system and neurotransmitter
The R2SN gradient in the subcortex and DMN shows significant alterations in AD

To identify these alterations in the R2SN gradient associated with AD, we computed the top 50 brain regions (about 20% of brain regions) exhibiting substantial differences between individuals with AD and NC (table S1). Of these regions, 21 were classified as belonging to the subcortex, primarily encompassing the basal ganglia (BG), hippocampus, and amygdala. Furthermore, nine regions were identified within the DMN, predominantly involving the subregions of the superior temporal gyrus (STG), such as STG_R_6_6, as well as the subregions of the medial temporal gyrus (MTG) specifically denoted as MTG_L_4_2 and MTG_R_4_4 (Fig. 3A and table S1). We also calculated the Pearson correlation of the spatial similarity between the *t*-maps from different sites. The results exhibited significant correlations between the *t*-maps for AD versus NC at each site and those from other sites, indicating that the altered brain gradient in AD is consistent and robust across different datasets (fig. S4).

Table 1. The summary of participants in this study.			
Dataset	Group	Age (year)	Sex (M/F)
HCP	Young NC (1113)	28.80 ± 3.70	507/606
Cam-Can	Cross-sectional adult life-span sample (652)	54.28 ± 18.65	322/330
ADNI	NC (605)	73.47 ± 6.16	279/326
	AD (283)	74.91 ± 7.70	152/131
EDSD	NC (141)	69.81 ± 5.77	69/72
	AD (124)	72.79 ± 7.52	57/67
NACC	NC (2042)	65.07 ± 10.71	658/1384
	AD (963)	71.78 ± 8.33	489/474
MCADI	NC (336)	64.71 ± 8.86	148/188
	AD (399)	69.23 ± 9.23	154/245
NIFD	NC (124)	64.37 ± 7.86	55/69
	BvFTD (56)	62.36 ± 7.01	23/33
	SV (37)	64.16 ± 6.14	13/24
	PNFA (36)	69.11 ± 7.43	17/19
PPMI	NC (87)	59.90 ± 11.01	57/30
	PD (181)	62.12 ± 9.45	117/64
	GenCohort PD (14)	60.07 ± 10.96	8/6
	SWEDD (37)	62.19 ± 10.13	23/14
XW_VaD	NC (24)	59.42 ± 6.29	4/20
	VaD (69)	65.87 ± 7.80	43/26

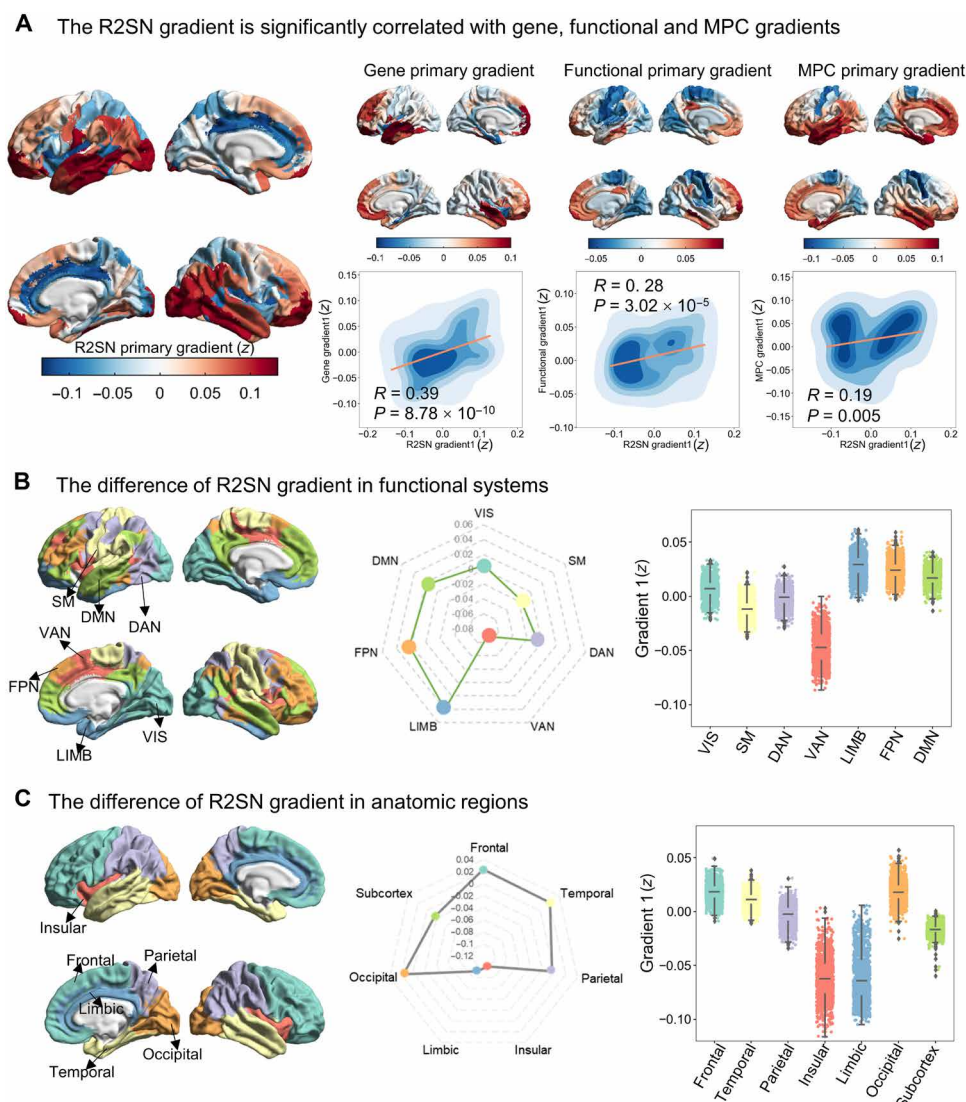


Fig. 2. The R2SN gradient in young adults. (A) The correlation between the R2SN gradient and gene gradients, functional gradients, and MPC gradients; (B and C) the distribution of the R2SN gradient in functional systems and anatomic regions. VIS, visual network; SM, somatomotor network; DAN, dorsal attention network; VAN, ventral attention network; LIMB, limbic network; DMN, default mode network.

However, a few sites with small samples, such as site 5 ($n = 35$), site 7 ($n = 39$), site 15 ($n = 32$), and site 17 ($n = 30$), displayed low reproducibility in the differences in brain gradients between AD and NC with other sites. This finding underscores the value of meta-analysis in comprehensively assessing alterations across different sites. Notably, it is intriguing that the altered pattern of the R2SN gradient in MCI shows a robust spatial resemblance to that in AD ($R = 0.86$ and $P < 0.001$) (fig. S4B), although it is less pronounced in MCI than in AD (fig. S3C). Consequently, this finding highlights that the gradient alteration within the AD continuum represents a continuous process, with changes occurring in MCI and intensifying in AD (fig. S5).

The R2SN gradient of AD was related to the biological process

We also explored the relationship between gene expression and the altered pattern of the R2SN gradient in AD. The findings revealed a statistically significant and positive correlation ($R = 0.39$

and $P = 3.21 \times 10^{-10}$) between the first partial least squares element (PLS1) and the observed altered gradient in AD (Fig. 3A). A gene enrichment analysis was conducted to gain further insights into the genetic basis of this altered gradient in AD, using the top 1000 genes ranked by PLS weight. Notably, this analysis highlighted gene terms associated with the neuronal system (R-HSA-112316, $P < 1 \times 10^{-13}$), the modulation of chemical synaptic transmission (GO:0050804, $P < 1 \times 10^{-10}$), regulation of monoatomic ion transport (GO:0043269, $P < 1 \times 10^{-10}$), and synaptic signaling (GO:0099536, $P < 1 \times 10^{-7}$), to particularly significant with the observed modified gradient in AD (Fig. 3B and table S2). This network visualization methodically elucidates the functional associations and emergent patterns among the enriched terms, offering insights into the biological processes implicated in altered gradients in AD. However, it is imperative to acknowledge that the results presented herein constitute preliminary evidence, due

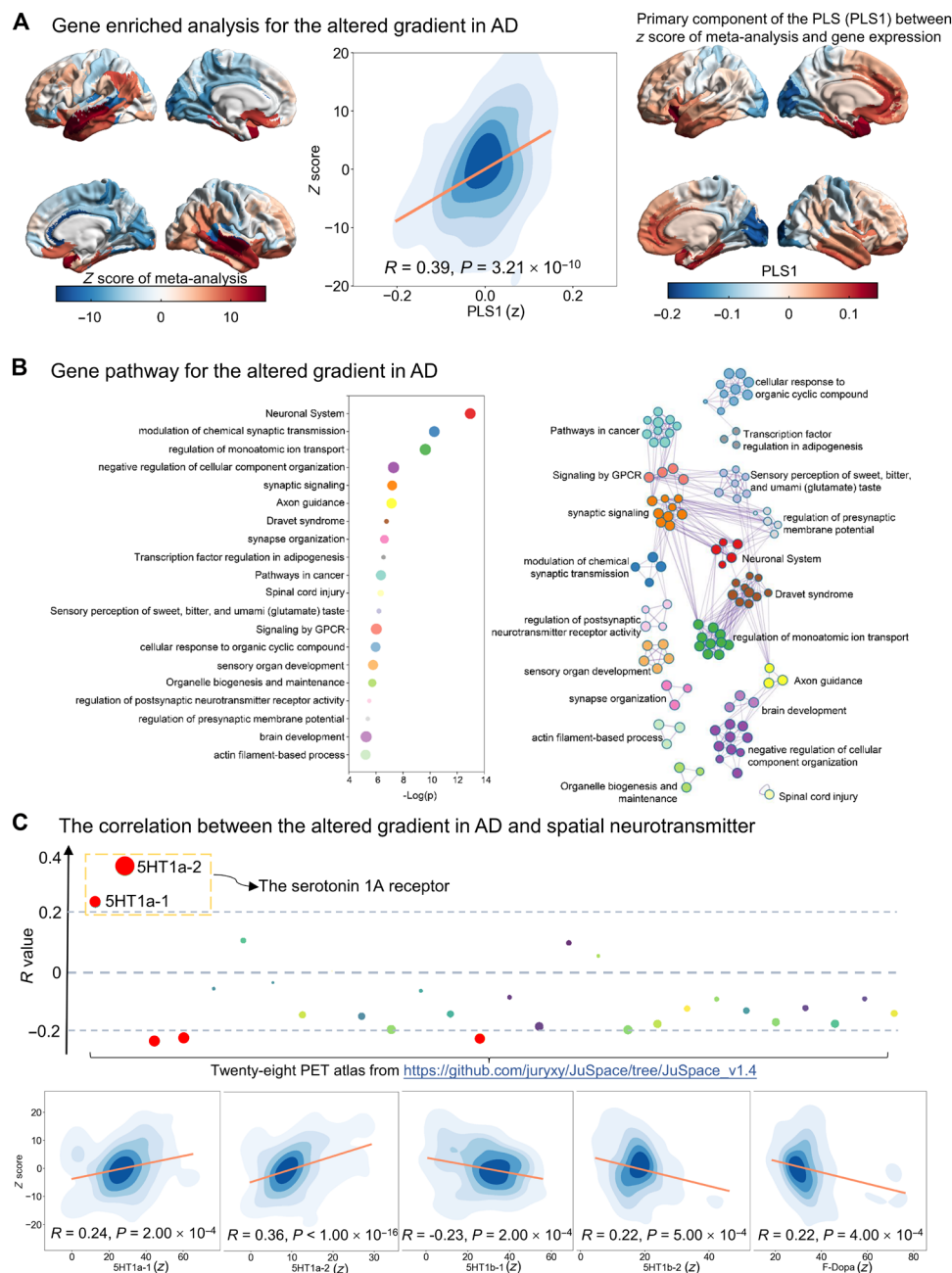


Fig. 3. The altered R2SN gradient in AD. (A) The altered pattern of the R2SN gradient in AD and the primary PLS component (PLS1) for gene expression. (B) The enriched gene terms and gene pathway for the altered pattern of the R2SN gradient in AD; it illustrates both intracluster and intercluster connections, delineating up to 10 terms per cluster. Each cluster is differentiated by a distinct color code, facilitating straightforward identification and analysis. (C) The distribution of the correlation coefficient between the altered pattern of the R2SN gradient in AD and spatial neurotransmitter maps. The vertical axis was the z score of the meta-analysis in AD versus NC and the horizontal axis was the value of the different regional neurophysiological characteristics. Of these, 28 PET/SPET atlas were obtained from https://github.com/juryxy/JuSpace/tree/JuSpace_v1.4.

to the lack of actual genetic data from participants. While this method is primarily based on reasonable assumptions supported by previous studies, further empirical validation is necessary.

The R2SN gradient of AD mirrors the neurotransmitter changes in the brain

To further explore the molecular mechanism of the modified R2SN gradient observed in AD, we investigated the spatial correlation between the 28 positron emission tomography (PET)/single-photon

emission tomography (SPET) receptor maps and the altered gradient in AD. Notably, our analysis identified the neurotransmitter system, specifically the receptors 5-hydroxytryptamine (5-HT): 5HT1A-1 and 5HT1A-2, as exhibiting the most significant correlation with the R2SN gradient in AD compared to the other receptor maps (Fig. 3C and table S3). Those convergent findings support the hypothesis that the neurotransmitter system may serve as the key determinant driving the observed modifications in the gradient associated with AD.

R2SN gradient shows a specific pattern in AD compared with aging and other neurodegenerative diseases

We conducted a comprehensive investigation into the dynamic changes of the R2SN gradient across the human life span using the Cam-Can dataset. We initially identified widespread brain regions that exhibit significant associations with age (Fig. 4A). Specifically, the differences in the brain gradient between older and younger individuals are consistent but more pronounced than those between middle-aged and younger individuals (Fig. 4, B and C). In contrast to AD, this distribution primarily encompasses both the unimodal region (including VIS, SM, and DAN) and subcortical areas (Fig. 5B and table S4).

Furthermore, the study elucidated AD's distinctive alteration in the R2SN gradient relative to other neurodegenerative diseases.

Specifically, the FTD, comprising behavioral variant (BV), semantic variant (SV), and progressive nonfluent aphasia (PNFA), exhibited alterations mainly in DMN and LIMB, while PD and GPD portrayed alterations primarily in LIMB. Last, the R2SN gradient of vascular dementia (VaD) was significantly impaired in SM (Fig. 5C). The findings offer insights into the underlying dysfunction of the brain connectome associated with specific clinical phenotypes across neurodegenerative disorders.

The alteration of the R2SN gradient shows a transcriptomic similarity in multiple neurodegenerative diseases

Our study revealed an enrichment of the Neuronal System (R-HSA-112316, $P < 1 \times 10^{-8}$) in both neurodegenerative diseases and normal aging. We observed a nearly universal modulation of the

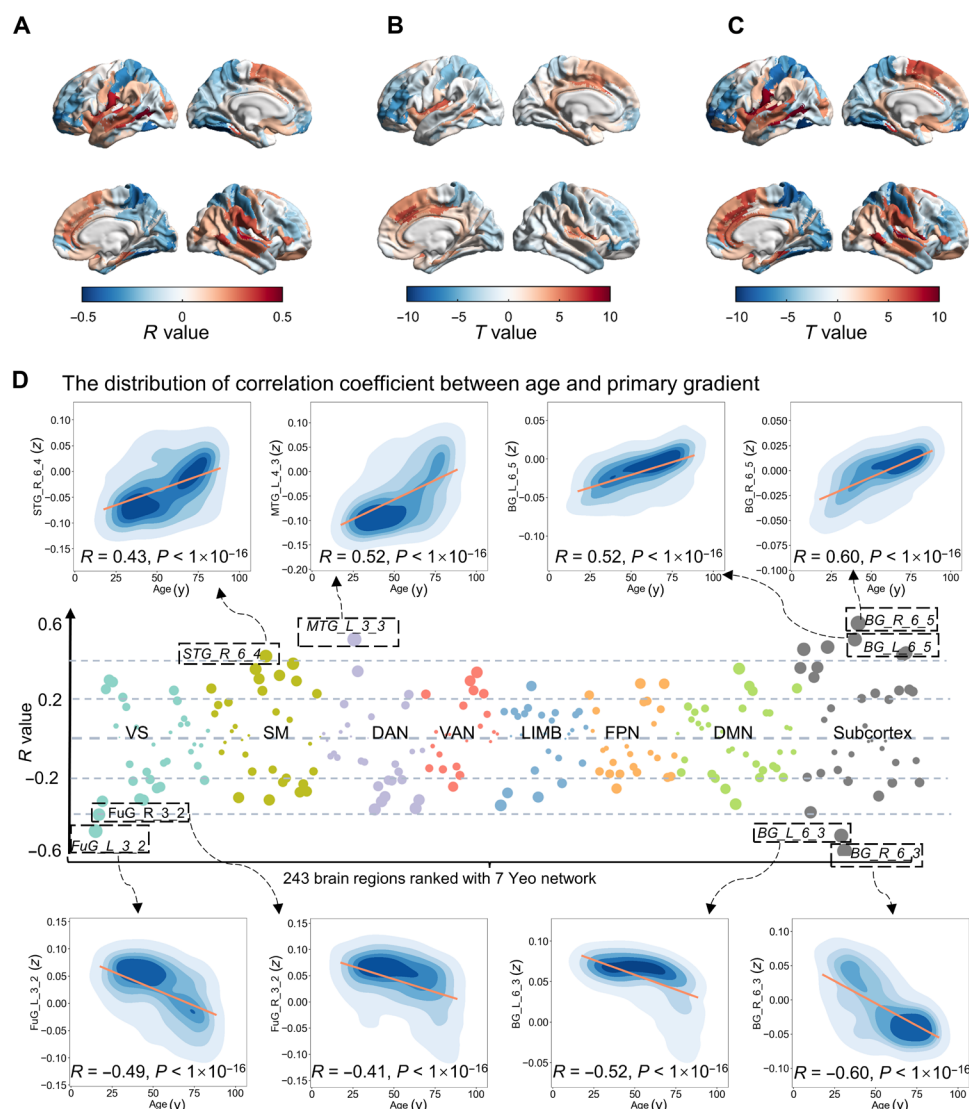


Fig. 4. The altered pattern of the R2SN gradient in aging. (A) The correlation between R2SN gradient and age; (B) the difference of gradient between middle age (age from 40 to 60) versus young (age from 20 to 40); (C) the difference of gradient between older (age from 60 to 80) versus young; (D) the distribution of the correlation coefficient and the typical regions between the R2SN gradient and age. The vertical axis of the kernel density map was the gradient value of different regions. STG_R_6_4, right superior temporal gyrus, caudal area 22; MTG_L_4_3, left middle temporal gyrus, dorsolateral area 37; FuG_R_3_2, right fusiform gyrus, medioventral area 37; BG_R_6_5, right BG, dorsal caudate; BG_L_6_5, left BG, dorsal caudate; BG_R_6_3, right BG, nucleus accumbens; BG_L_6_3, left BG, nucleus accumbens. More detailed information can be found in table S21.

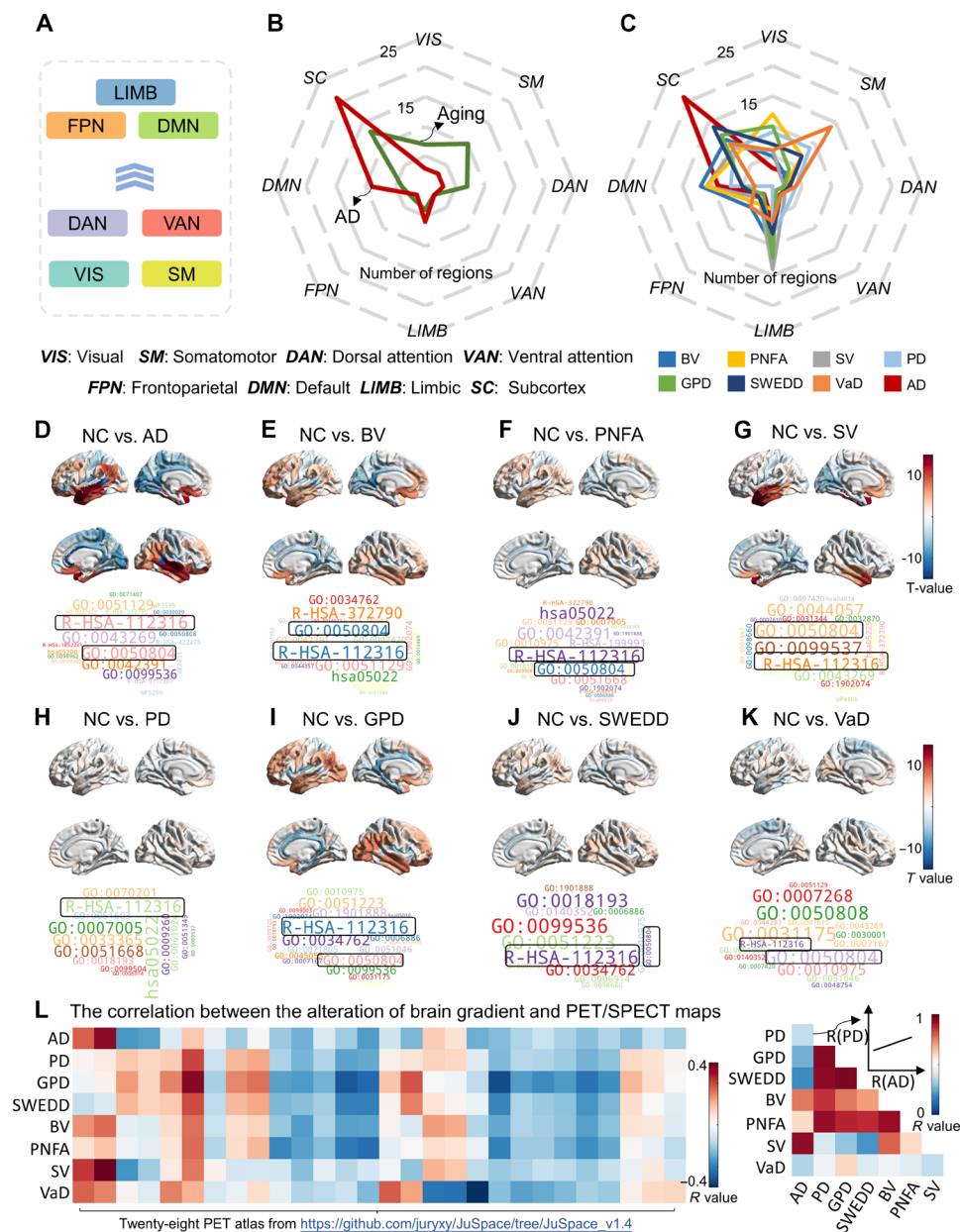


Fig. 5. The modified R2SN gradient in various neurodegenerative conditions. (A) The hierarchical structure and organization of the brain; (B) the number of brain regions in each Yeo network among the top 50 brain regions deemed significant in aging and AD; (C) the number of brain regions in each Yeo network among the top 50 brain regions of significance in AD and other neurodegenerative disorders; (D) to (K) the altered pattern and their related gene terms for each disease; (L) the correlation between the alteration of brain gradient and 28 PET/SPET maps in eight neurodegenerative diseases; and the similarity between the *R* value of brain gradient alteration and PET/SPET in different diseases. BV, behavioral variant FTD; SV, semantic variant; PNFA, progressive nonfluent aphasia; PD, Parkinson's disease; GPD, genetic PD; SWEDD, scans without evidence for dopaminergic deficit; VaD, vascular dementia; R-HSA-112316, neuronal system; GO:0050804, the modulation of chemical synaptic transmission.

chemical synaptic transmission pathway (GO:0050804, $P < 1 \times 10^{-7}$) across these disorders, indicating a gradient-related transcriptional similarity among neurodegenerative diseases (Fig. 5, D to K) (tables S2 and S5 to S11). Conversely, inorganic ion transmembrane transport (GO:0098660, $P < 1 \times 10^{-15}$) showed the most significant correlation in aging (table S12). These findings suggest that the neuronal system is a common target in aging and neurodegenerative diseases, influenced by distinct pathways (table S13).

The chemical synaptic transmission showed specific dysfunction between AD and other neurodegenerative diseases

In our investigation into the biomechanisms of neurodegenerative diseases, we analyzed 28 PET/SPECT neuropathophysiological maps, uncovering distinct synaptic alterations. Specifically, in AD, 5HT1A and 5HT1B receptors were key synaptic contributors. Notably, it showed unique associations with different receptors and transporters,

i.e., behavioral variant FTD (bvFTD) was associated with 5HT1A and D2 receptors, PD with 5-HT2A receptors and the dopamine transporter, and VaD with brain noradrenaline and μ -opioid receptors. The findings indicate that while chemical synaptic transmission is a common feature enriched across various neurodegenerative diseases, each disease is associated with unique dysfunctional neurophysiological characteristics (Fig. 5L, Table 2, and table S3).

DISCUSSION

This study establishes the R2SN gradient as a neuroimaging measure for the detailed characterization of the brain's topographical structure. It is a biologically plausible marker for identifying specific hierarchical structural dysfunctions in AD. Our findings reveal transcriptomic similarities in chemical synaptic transmission across neurodegenerative diseases, with distinctive synaptic patterns in AD. These insights are critical for understanding the complex overlaps and distinct clinical phenotypes in these disorders (Fig. 6).

Previous studies have extensively examined the anomalies observed in the functional gradient architecture. Notably, these investigations have focused on its dysfunctions within the context of major depressive disorder (29) and schizophrenia (15), as well as its perturbations during development (30). Instead, the structural gradient derived from the morphological similarity has not been studied well yet (26, 31). The biological meaning of structural covariance remains controversial, but it appears to reflect developmental coordination or synchronized maturation between brain areas (6). A recent study also showed the feasibility of capturing a brain hierarchical structure with a brain structural similarity network (26). Consistently aligning with expectations, this study presents a systematic characterization of the topographical structure of the brain in the context of young adults. It provides additional evidence that the brain structural similarity network gradient can capture the organizational patterns within the brain.

We observed spatial heterogeneity within Yeo networks (32), with the R2SN gradient manifesting lower values in the unimodal region, encompassing the VAN, SM, DAN, and VIS, while higher values were localized in the LIMB, FPN, and DMN. This spatial continuum provides compelling evidence for a hierarchical organization ranging from unimodal to transmodal regions, shedding light on the brain module's functional architecture, segregation, and integration (8). Besides, those findings demonstrated that the R2SN gradient can capture the intricate topographical structure of the brain. This study

constitutes an essential extension of prior investigations into the hierarchical organization of the brain, particularly concerning the brain structural similarity network (26).

The robust pattern of alteration observed in the brain gradient in AD holds importance in elucidating the underlying cognitive decline mechanisms associated with AD. It is imperative to estimate this alteration pattern using a multisite dataset rather than relying solely on a single dataset (33). A data-driven meta-analysis based on unified processed features represents a powerful approach that not only overcomes data heterogeneity but also enables more precise estimations and facilitates regional comparisons, thereby advancing our understanding of complex neurobiological phenomena (34–36). We systematically evaluated the alteration of the brain gradient in AD based on a large and multisite dataset under the meta-analysis framework. The brain gradient in the subcortex and DMN shows the most significant difference between AD and NC, primarily encompassing the BG, hippocampus, amygdala, and parts of STG and MTG. Previous studies have reported the structural profiles of those regions (17). Natural aging primarily impacts the primary cortex and subcortex regions, leading to dispersion and average value changes. The modified pattern observed in the R2SN gradient during normal aging does not indicate a decline in any singular function; instead, it suggests a generalized alteration without pinpointing a specific function impairment. Conversely, the gradient of DMN was significantly altered in AD, which reveals the brain mechanism of the cognitive ability decline in AD. Therefore, this result potentially links connectome gradient disruption and cognitive impairments in AD. Moreover, it offers evidence of the underlying reasons for the distinctive functional impairments observed in AD and the natural aging process.

We also explore the biological mechanism for those alterations based on the radiogenomics framework. Given the inherent difficulties in obtaining the gene expression across human brain tissue, we have used an “imaging transcriptomics” approach based on the Allen Brain dataset. This methodology enables us to decode the molecular correlations of the neuroimaging phenotype, establishing connections between spatial variations in both the transcriptomic and imaging domains. Although this approach may be considered somewhat indirect, it has the potential to unveil biologically plausible mechanisms, a methodology that has gained prominence in recent studies in the field (37–39). Gene enrichment analysis of the altered R2SN gradient in AD yielded compelling results, indicating significant associations with gene terms relevant to the neural

Table 2. The neurotransmitter terms that showed the most significant correlation to each neurodegenerative disease.	
Disease	Neurotransmitter terms
AD	5-HT1A receptor, 5-HT1B receptor
bvFTD	5-HT1A receptor, D2 receptor
SV	5-HT1A receptor, 5-HT1B receptor
PNFA	Fluorodopa, dopamine transporter
PD	5-HT2A receptor, dopamine transporter
GPD	5-HT2A receptor, 5-HTT receptor
SWEED	Fluorodopa, dopamine transporter
VaD	Brain noradrenaline, μ -opioid receptors

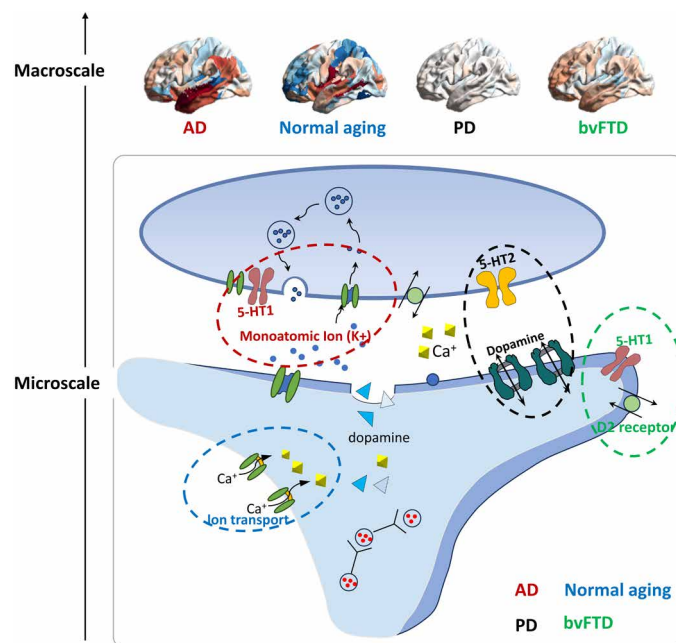


Fig. 6. The schematic diagram of the biomechanism of AD and other neurodegenerative diseases, ranging from the microscale to the macroscale.

system and neurotransmitter function. The enriched gene terms encompassed synaptic-related processes, which play a crucial role in AD (40, 41). It should be noted that these results should be considered uncertain conclusions rather than conclusive evidence, given that the genetic data are achieved from six healthy subjects in the Allen Brain dataset.

The R2SN diverges fundamentally from conventional functional brain networks, embodying structural similarity networks based on regional textural similarity (42). R2SN gradients are a foundational organizing principle and topographical structure within the macroscale structural connectome, independent of conventional neural system delineations (26). Intriguingly, gene enrichment analyses revealed that the altered R2SN gradients in AD correlated with the specific dysfunction of neuronal systems and neurotransmitter pathways (43). The observed correlations between brain gradients and neurotransmitter profiles suggest a potential explanation for the connectome dysfunction: Neuronal degeneration may disrupt standard patterns of neurotransmitter transport across membranes, altering regional textures and contributing to the macroscale connectome's dysfunction, consistent with established findings (44). These results further corroborate the characterization of AD as a disconnection syndrome marked by neuronal system dysfunction (45–47). In addition, the study proposes that macroscale network gradients could serve as a neuroimaging marker for biological dysfunction, potentially bridging macroscale and microscale assessments. This perspective introduces a potential hypothesis for understanding the biological foundations and mechanisms underlying structural similarity networks: The anatomical covariation network integrates diverse anatomical metrics to decode microscale brain data, thereby capturing synchronized developmental or degenerative patterns across paired brain regions (6).

Furthermore, the correlation analysis between PET/SPET receptor maps and the altered R2SN gradient in AD demonstrated strong associations, in particular within the neurotransmitter system (48, 49).

Synapses are the connections between nerve cells (neurons) that allow them to communicate with each other through the transmission of neurotransmitters (50). One of the hallmarks of AD is the accumulation of abnormal protein deposits in the brain, including beta-amyloid plaques and tau tangles. These deposits can interfere with synaptic function and damage synapses (51). Neurotransmitter imbalances further affect communication between neurons, ultimately leading to the deterioration of cognitive abilities (52). Assessing regional neurotransmitter functions within the brain *in vivo* poses a substantial challenge. The correlation findings demonstrated that alterations in the R2SN gradient may be linked to specific changes in neurotransmitter activity or receptor expression levels. These findings suggest that the alteration of the R2SN gradient provides promising insight to reveal the brain biomechanism in cognitive decline. Further investigations into the relationship between brain gradient and specific genes, molecular pathways, and neurotransmitter systems can provide a deeper understanding of the pathophysiology of AD and potentially guide the development of targeted therapeutic interventions (53).

In our investigation, we directed our attention toward the distinct patterns of altered gradients observed in AD, which deviate from those associated with normal aging and other neurodegenerative disorders. We identified distinct regional disparities in the modified pattern of the R2SN gradient in AD within DMN and subcortex. Three subtypes of FTD showed a similar altered pattern, exhibiting alterations primarily within the DMN and limbic systems. Besides, PD and GPD displayed alterations primarily in the limbic system. Last, the R2SN gradient impairment in VaD was primarily observed within the SM. The outcomes of our study provide compelling evidence of shared alterations across various brain regions, notably the DMN, in the context of multiple neurodegenerative disorders. In addition, our investigation revealed the presence of distinctive and disorder-specific patterns of altered gradients, emphasizing the uniqueness inherent to each disease.

Furthermore, this study finds that the neuronal system is linked to the altered brain gradient in neurodegenerative diseases and normal aging. However, the modulation of chemical synaptic transmission correlates with this altered gradient in neurodegenerative diseases (54). In contrast, inorganic ion transmembrane transport shows the most significant correlation in normal aging (55), suggesting that ion transport may play a primary role in the aging process, as opposed to chemical synaptic transmission (56). Brain cells can be categorized into neurons and non-neuronal cells, with neurons receiving substantial support from other cell types to maintain tissue homeostasis (18). Specifically, the inorganic ions play crucial roles in cellular functions; for example, potassium ions are involved in microglial cell activity (13). Thus, while inorganic ion transmembrane transport does not directly cause neuronal dysfunction, it affects non-neuronal cells, which subsequently affects the activity of neuronal systems, rather than a specific dysfunction. This interaction aligns with the broad yet mild decline in function observed in aging and aligned with a previous study (18). In contrast, specific dysfunctions in neurotransmitter systems can result in marked impairments in higher cognitive functions, corresponding with the distinct clinical phenotypes observed in neurodegenerative disorders (57). These results elucidate that neurosystem alterations could serve as a shared target for both aging and neurodegenerative disorders, meanwhile with distinct pathways potentially leading to specific downstream effects.

Synaptic dysfunction has been suggested to be the early event of several neurodegenerative disorders (58). Besides, neuro-inflammatory and neurodegenerative conditions are often associated with cognitive decline due to neural injury and disruption of synaptic connections (59). Two typical patterns were obtained in comparison to the chemical neurotransmitter transmission among neurodegenerative disorders: (i) The results revealed transcriptomic similarities among various neurodegenerative disorders; for instance, 5HT1A receptors are correlated with AD, bvFTD, SV, and PD. This kind of association suggests that common microscale functional processes might exist across different neurodegenerative disorders, consistent with overlapping macroscale neuropathology and clinical phenotypes (57). (ii) The results also indicated distinct transcriptomic profiles among neurodegenerative disorders. For example, in AD, the 5HT1A and 5HT1B receptors were identified as vital synaptic contributors (60–62), while bvFTD was associated with 5HT1A and D2 receptors (63). PD was linked to 5-HT2A receptors and the dopamine transporter (64). These findings emphasize that while chemical synaptic transmission is a common feature across neurodegenerative diseases, each disease exhibits unique dysfunctional neurophysiological characteristics. In addition, the results suggest that it is not a single chemical synaptic transmission but rather a combination of multiple synaptic transmissions that may collectively influence the variability in clinical phenotypes and neuropathology observed in neurodegenerative diseases. The brain has numerous potential drug targets due to its extensive neuro systems (65). Notably, certain systems, such as the 5-HT (61, 62) and dopaminergic systems (64, 66), have been identified as potential focal points for disease-modifying therapies. This study suggests that the interplay of multiple synaptic transmissions may play crucial roles in neurodegenerative diseases. Consequently, therapeutic targets should consider the comprehensive network of neuro systems in the brain, rather than focusing solely on traditional single functions and systems. It is also important to acknowledge that these speculative conclusions must be validated through further animal experiments.

Limitation

It is essential to acknowledge the limitations of our study. First, it is important to note that the above results from the genetics enrichment analysis are introduced to explain and tentatively explore the biological basis of the alteration of brain gradient, and depend on the genetic data derived from only six healthy subjects in the Allen Brain dataset. The sample size of non-AD neurodegenerative diseases was relatively small, and future studies with more extensive and diverse populations are necessary to validate our findings. Longitudinal studies are also warranted to investigate the temporal dynamics of the R2SN gradient and its relationship with disease progression. In addition, further research is needed to explore the potential of the R2SN gradient in other neurodegenerative disorders and to elucidate the underlying mechanisms driving the observed alterations. Furthermore, the gradient of the cerebellum should be explored further in future studies.

MATERIALS AND METHODS

Participants and clinical assessments

This multicohort study encompassed nine independent datasets, consisting of a normal young dataset [Human Connectome Project dataset (HCP), $n = 1113$], an aging dataset (Cam-Can, $n = 652$), four multisite AD cohorts including the Alzheimer's Disease Neuroimaging Initiative [ADNI; three sites (ADNI1&GO, ADNI2, and ADNI3) with $n = 888$], the European DTI Study on Dementia (EDSD; six sites with $n = 265$), the National Alzheimer's Coordinating Center (NACC; 12 sites with $n = 3005$), and in-house Multi-Center Alzheimer Disease Imaging Consortium dataset (MCADI; eight sites with $n = 735$). Besides, to further explore the altered gradient pattern in the AD continuum, the MCI of those datasets was also included in the present study (tables S14 to S16).

Furthermore, three datasets with other neurodegenerative disorders were included. The Neuroimaging in Frontotemporal Dementia (NIFD) dataset consisted of individuals with various conditions, including NCs ($n = 124$), those diagnosed with bvFTD ($n = 56$), SV ($n = 37$), and PNFA ($n = 36$). Similarly, the Parkinson's Progression Markers Initiative (PPMI) dataset comprised NCs ($n = 87$), those diagnosed with PD ($n = 181$), individuals with genetic PD (GPD, $n = 14$), and individuals with scans showing no evidence of dopaminergic deficit (SWEDD; $n = 37$). In addition, the VaD_XW dataset included NCs ($n = 24$) and those diagnosed with VaD ($n = 69$) (table S17).

This study received ethical approval from the Biomedical Ethical Review Committee of Beijing University of Posts and Telecommunications (approval no. 202402016). The participants included in the MCADI dataset were approved by the respective Medical Ethics Committees of local hospitals in China, and each participant provided written informed consent. Additional datasets were obtained from publicly accessible sources, with all necessary ethical approvals in place. Detailed information can be found in S01 in the Supplementary Materials.

Data processing and gradient computing

For each participant, the T1-weighted image was registered to Montreal Neurological Institute standard space implemented Advanced Normalization Tools with the "SyN" method after denoising and N4 correction. Subsequently, to obtain the regional morphology profile, a series of radiomics features (14 intensity features and 33 texture features) of each brain region defined as Brainnetome atlas (67) were extracted (36, 68).

The redundancy feature was defined as features with a high correlation with other features ($R > 0.9$). As a result, a final 25 radiomics feature was obtained for further analysis. A detailed pipeline and definition of those radiomics features can be found in tables S18 to S20. Then, a standard min-max method was introduced to normalize the radiomics feature among different brain regions in an individual. The individual R2SN was created by mapping each individual's radiomics features into a pairwise interregional Pearson correlations matrix after feature normalization. Essentially, a 246×246 connection matrix was generated for each subject, with nodes representing regions from the Brainnetome atlas (table S21) and edges calculated on the basis of Pearson correlations between interregional radiomics features (<https://github.com/YongLiuLab> and <https://zenodo.org/records/13160573>) (details can be found in S02 in the Supplementary Materials).

The topological organization of the brain structure was evaluated on the basis of R2SN connectomes using the diffusion map-embedding technique (69). In brief, diffusion map-embedding was a nonlinear dimensionality reduction technique that operates by simulating a diffusion process on the data. This approach transforms the Euclidean distances between datapoints into probabilities of state transitions, thus capturing the intrinsic structure and dynamics of the data. The diffusion map-embedding method is particularly adept at revealing the manifold's underlying geometry and is effectively used to characterize the topographical structure of complex networks (8). Diffusion map-embedding provides a lower-dimensional representation that preserves the essential features of the data manifold by using eigenvalue decomposition on the diffusion operator, thereby elucidating connectivity patterns and structural organization within the network. This process was implemented using the BrainSpace toolkit, available at <https://brainspace.readthedocs.io/en/latest/>. Here, the group-level connectivity matrix, constructed on the basis of the HCP dataset, was used as the foundational template for calculating gradients in the current study.

The R2SN gradient in young adults

The first component, referred to as gradient 1, was the principal gradient capable of capturing the topological organization of morphological profiles across the entire brain (26). To explore the biological basis of the R2SN gradient, we examined its associations with the functional connectome profile, genetics gradient, and MPC gradients. Here, the R2SN gradient was computed at the group level using the HCP dataset, resulting in a 246×1 vector. The functional and MPC gradients can be accessed at <https://github.com/MICA-MNI/micaopen/tree/master/MPC>. The gene similarity network was derived by calculating regional gene expression similarity based on the Allen human brain atlas datasets (<https://human.brain-map.org/>), and the genetics gradient was computed using the same protocol as the R2SN gradient.

To further explore the robustness of the R2SN gradients for different machine sequence parameters, two traveling datasets [dataset1, three healthy traveling subjects with 10 sites (70); and dataset2, nine healthy participants with 12 sites (71)] were included in the present study to estimate the reproducibility of R2SN gradient (details can be found in Supplementary Materials S01). Here, the ICC served as a quantitative index to estimate the reproducibility of R2SN gradients. The ICC has a value between 0 and 1; $ICC = 0$ indicates no reproducibility, and $ICC = 1$ indicates absolute reproducibility (72).

Furthermore, we investigated the distribution of the R2SN gradient across different functional systems and anatomical parcellations at both the group and individual levels. Specifically, the functional

systems were defined as the Yeo networks (32), and the definition of brain lobes was according to the Brainnetome atlas.

The alteration of the R2SN gradient in AD and its relationship between genetics and neuropathophysiological maps

A meta-analysis framework was used to comprehensively investigate the alterations of the R2SN gradient associated with AD, integrating gradient differences across four multisite datasets (ADNI, EDS, NACC, and MCADI). In this framework, the initial step involved computing the gradient difference between the AD and NC groups within each site using a two-sample, two-sided t test. It should be noted that the covariates considered in this study were limited to age and gender. Other clinical information, such as APOE4 status, was not included for the following reasons: (i) APOE4 status was only available for a subset of the ADNI dataset; and (ii) APOE4 is a known high-risk factor for AD and is associated with abnormal brain structural patterns, which could independently influence brain gradient changes. Subsequently, the Liptak-Stouffer z score method (73) was applied to aggregate the results from different sites (34–36). In brief, the P value for each brain gradient at the i th site was transformed into its corresponding z_i score using the formula: $\Phi^{-1}(1 - p_i)$, where Φ^{-1} represents the inverse of the standard normal cumulative distribution function. Subsequently, a combined z score for each feature was obtained using the Liptak-Stouffer formula as follows

$$z = \frac{\sum_{i=1}^k w_i z_i}{\sqrt{\sum_{i=1}^k w_i^2}}$$

In this context, w_i denotes the inverse of the variance in z_i , serving as a relative measure of statistical power compared to other datasets. Under the null hypothesis, the Z scores are anticipated to follow a standard normal distribution. This methodology was used to compute the P values based on the corresponding Z scores to test for group differences between the NC and AD groups. Simultaneously, we applied the same meta-analysis framework to explore the significant alterations in the R2SN gradient linked to MCI.

The gene expression data were obtained from the Allen atlas (<http://human.brain-map.org/>) (74) and subsequently aligned to the Brainnetome atlas using the “abagen” toolkit (<https://github.com/rmarkello/abagen>). We used the PLS correlation analysis to effectively rank 15,633 genes based on their corresponding weight values, allowing for further investigation of their association with the R2SN gradient. Gene set enrichment analysis was also performed on the top 1000 genes using the Metascape platform (<https://metascape.org/gp/index.html#/main/step1>) (75).

We used PET/SPECT maps of 28 neuropathophysiological features from unrelated control groups were incorporated (https://github.com/juryxy/JuSpace/tree/JuSpace_v1.4/JuSpace_v1.4) to assess the potential associations between the altered pattern of the R2SN gradient in AD and the neuropathophysiological changes. Pearson correlation was calculated between the Z value of the altered map in AD and regional mean PET/SPECT values, allowing for exploring potential associations between the R2SN gradient and neuropathophysiological changes.

The altered patterns of R2SN gradient in aging and other neurodegenerative disorders

Individual gradients were computed using the Cam-Can dataset (comprising participants aged between 18 and 88) to investigate the altered gradient patterns along the life span. Pearson correlation coefficient between age and the regional gradient was initially calculated. To further examine the impact of age on the gradient, the Cam-Can dataset was subdivided into three groups: young adults (ages 20 to 40), middle-aged individuals (ages 40 to 60), and older adults (ages 60 to 80). Subsequently, the differences in the gradient between young adults and middle-aged individuals, as well as between young adults and older adults, were calculated.

Furthermore, an analysis was conducted to examine the altered pattern of the R2SN gradient in other neurodegenerative disorders and its association with specific biological pathways. For each dataset, the altered pattern of the R2SN gradient was computed by comparing each disease group to its corresponding NC group, of which age and gender were included as covariates. The methodology used for gene set enrichment analysis in each neurodegenerative disorder followed the same approach as the analysis performed in AD.

Supplementary Materials

This PDF file includes:

Supplementary Text S1 to S3

Figs. S1 to S5

Tables S1 to S21

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